GITAM Deemed to be University

Biotechnology | Course Code: BT301

Genetic Engineering & Recombinant DNA - Laboratory Manual & Practical Gu

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Syllabus Focus: BT301 Laboratory Practical Manual & Viva Voce Handbook

Summary: Structured laboratory workbook detailing experiment objectives, baseline toolchains, e

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Molecular Tools & Restriction Enzymes

" Syllabus Scope: Type II restriction endonucleases, DNA ligases, Alkaline phosphatases, Reverse transcri

Unit 2: Cloning Vectors & Transformation

" Syllabus Scope: Plasmids pBR322, pUC19, Bacteriophage lambda, Cosmids, YAC, BAC, Competent cell prepara

BT301: Genetic Engineering & Recombinant DNA - Advanced Topics & Assessment

Academic Track: Biotechnology - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Gene Amplification & Sequencing

" Polymerase Chain Reaction PCR variations RT-PCR, qPCR, Sanger chain termination sequencing, Next-Generation

Unit 4: Genome Editing & CRISPR-Cas9

" CRISPR-Cas9 target recognition, sgRNA design, Zinc Finger Nucleases, Recombinant therapeutics Insulin, Monoc

Unit 5: Expression Systems & Transgenic Organisms

" Prokaryotic vs Eukaryotic expression vectors, Inclusion bodies, Transgenic plants and animals, Bioethics and

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

Lab 1: Isolation of Plasmid DNA from Escherichia coli using Alkaline Lysis Method

Lab 2: Restriction Digestion of Lambda DNA using EcoRI and HindIII followed by Agarose Gel Electrophoresi

Lab 3: Polymerase Chain Reaction PCR Amplification of Target Gene Fragment with Primer Optimization

Lab 4: Preparation of Chemically Competent Bacterial Cells and Heat-Shock Transformation

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* T.A. Brown - Gene Cloning and DNA Analysis: An Introduction Wiley-Blackwell 7th Ed

* Bernard R. Glick - Molecular Biotechnology: Principles and Applications of Recombinant DNA ASM Press

Authorized NoteVault Educational Document " Verified for Registered University Students